

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Six Semester of B. Tech. (EE) Examination

May 2014

EE309.01 ~~Electrical~~ Power Utilization & Traction

Date: 13.05.2014, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Draw simplified speed time curves and Calculate V2 by quadrilateral speed time curve. [07]

Q - 2 (a) Write short note on system of track electrification. [07]

(b) Explain dynamic braking for D.C. series & shunt motor. [07]

OR

Q - 2 (a) Explain D.C. and A.C. Rheostatic braking for three phase induction motor. [07]

(b) Explain tractive effort for propulsion of train. [07]

Q - 3 (a) Derive Equation for Temperature rise of electric drive for Heating curve. [07]

(b) A motor has a heating time constant of 70 minutes and cooling time constant of 90 minutes when stationary. When run continuously on full load of 18.5kw, the total temperature rise is 40c. What load could the motor deliver for 10 minutes if this is followed by shut down period long enough for it to cool without the temperature rise exceeding 40c. if it is on intermittent load of 10 minutes on load and followed by 15 minutes of shut down, what is the maximum value of load which it could supply during the on-load period without maximum temperature rise exceeding 40c ? Assume losses to be proportional to square of the load. [07]

OR

Q - 3 (a) Derive flywheel calculation equation when motor load is increasing and motor load is decreasing. [07]

(b) An electric motor has load variation as given below: torque 240 Nm for 20 minutes, 140 Nm for 10 minutes, 300 Nm for 10 minutes, 200 Nm for 20 minutes. If the speed of the motor is 720 rpm, find the power rating of the motor. [02]

(c) A motor fitted with a flywheel supplies a load torque 1000 N-m for 2 sec. during no load flywheel regain its original speed. The motor torque is required to be limited to 500 N-m. Determine moment of inertia of flywheel. The no load speed of motor is 500 [05]

rpm & its full load slip is 10%.

SECTION – II

- Q - 4 (a) List the different types of resistance and arc welding and explain butt welding. [07]
- Q - 5 (a) Explain direct and indirect resistance heating and write the advantages of electric heating. [07]
- (b) Explain photometry and list different types of photometer heads. [05]
- (c) Explain basic types of reflection. [02]

OR

- Q - 5 (a) Explain direct and indirect arc furnaces with necessary diagram. [06]
- (b) Explain street lighting. [04]
- (c) Explain design of lighting schemes. [04]
- Q - 6 (a) Explain transition methods. [07]
- (b) A 200 tonne motor coach having 4 motors each developing 6,000 Nm torque during acceleration, start from rest. if the gradient is 30 in 1000, gear ratio 4, gear transmission efficiency 90%, wheel radius 45 cm, train resistance 50 N/tonne, addition of rotational inertia 10 % calculate the time taken to attain a speed of 50 kmph . if the line voltage is 3000 V dc and efficiency of motor 85% ,find the current during notching period. [07]

OR

- Q - 6 (a) Write the comparison between D.C. and A.C. traction system. [05]
- (b) 400 tonnes goods train is to be hauled by a locomotive up a gradient of 2% with acceleration of 1 kmphs. Coefficient of adhesion is 20 %, track resistance 40 N/tonne and effective rotating masses 10% of the dead weight. Find the weight of the locomotive and number of axles if the axle load is not to increase beyond 22 tonnes. [07]
- (c) Explain faraday's laws of electro deposition. [02]
